Pertemuan 4 : Docker Multi Container

1. Buat Dockerfile untuk Apache + PHP Web Server !
   Membuat image PHP Web Server berbasis Apache dengan modifikasi penambahan extension yang diperlukan untuk koneksi MySQL. Langkahnya adalah sebagai berikut:

   - Buat sebuah direktori proyek baru dan buka dengan VS-Code
   - Buat file dengan nama Dockerfile dan isi dengan konten sebagai berikut:

```
FROM php:8.2-apache
WORKDIR /var/www/html

RUN apt-get update && \
    apt-get install -y libpng-dev && \
    docker-php-ext-install pdo pdo_mysql gd

RUN docker-php-ext-install mysqli && docker-php-ext-enable mysqli

COPY . /var/www/html/

RUN chown -R www-data:www-data /var/www/html
RUN a2enmod rewrite

EXPOSE 80
CMD ["apache2-foreground"]
```

   - Buat file index.php

```
<?php
echo "Docker PHP Apache is running!";
```

   - Buka built in terminal VS-Code untuk menjalankan perintah build image:

     `docker build -t {username}/{nama_image} .`

     contoh :

     `docker build -t a71b/p4 .`

   - Run image tersebut:

```
docker run -v ./www:/var/www/html -p 8081:80
{username}/{nama_image}

   contoh :

docker run -v ./www:/var/www/html -p 8081:80 a71b/p4
```

2. **Membuat Docker Compose**

a. Buat folder proyek baru, contoh: latihan4

```
latihan4/
├── docker-compose.yml
├── Dockerfile
├── default.conf
└── src/
    ├── index.php
    ├── (file project lainnya)
```

b. Buat file Dockerfile dengan isi berikut:

```
FROM php:8.2-fpm
WORKDIR /var/www/html
RUN docker-php-ext-install mysqli pdo pdo_mysql
COPY ./src /var/www/html
```

c. Buat file default.conf dengan isi:

```
server {
    listen 80;
    server_name localhost;

    root /var/www/html;
    index index.php index.html;

    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }

    location ~ \.php$ {
        include fastcgi_params;
        fastcgi_pass php-fpm:9000;
        fastcgi_index index.php;
        fastcgi_param SCRIPT_FILENAME
$document_root$fastcgi_script_name;
```

```
    }

    location ~ /\.ht {
        deny all;
    }
}
```

d. Buat file `docker-compose.yml` dengan isi :

```
services:
  web:
    image: nginx:latest
    container_name: nginx_web
    ports:
      - "8081:80"
    restart: always
    volumes:
      - ./src:/var/www/html
      - ./default.conf:/etc/nginx/conf.d/default.conf
    depends_on:
      - php-fpm

  php-fpm:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: php_fpm
    restart: always
    volumes:
      - ./src:/var/www/html

  mysql-db:
    image: mysql:5.7
    container_name: mysql_db
    command: --default-authentication-
plugin=mysql_native_password
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: testdb
      MYSQL_USER: user
      MYSQL_PASSWORD: password
    ports:
      - "3308:3306"
```

```
  phpmyadmin:
    image: phpmyadmin:latest
    container_name: phpmyadmin_ui
    restart: always
    ports:
      - "8082:80"
    environment:
      PMA_ARBITRARY: 1
      PMA_HOST: mysql-db
    depends_on:
      - mysql-db
```

e. Buat file index.php di folder src dengan isi :

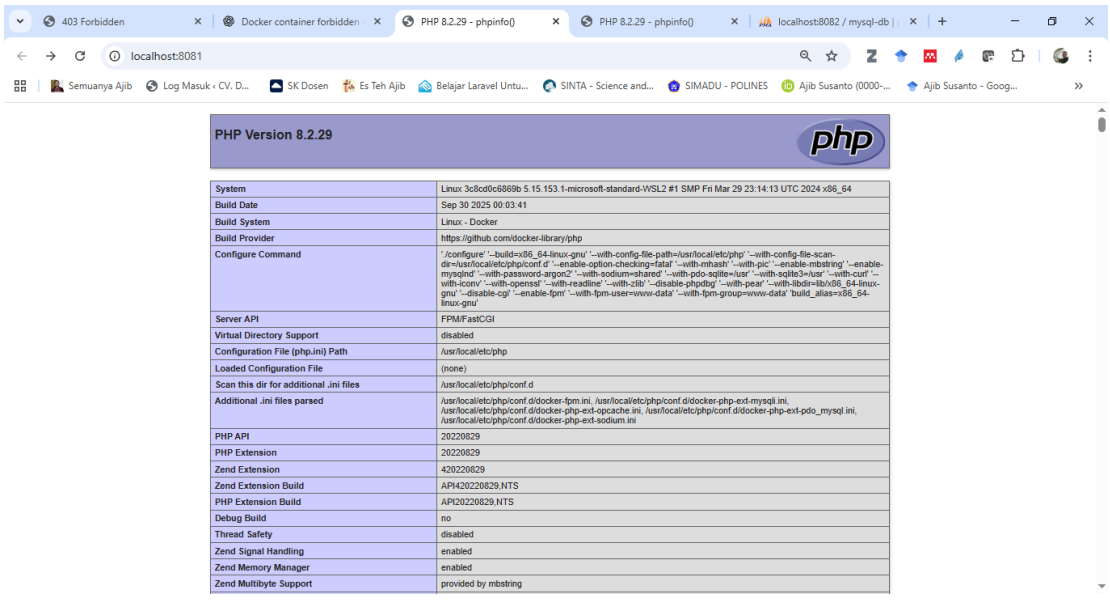
```
<?php
      phpinfo();
?>
```

f. Jalankan dan coba :

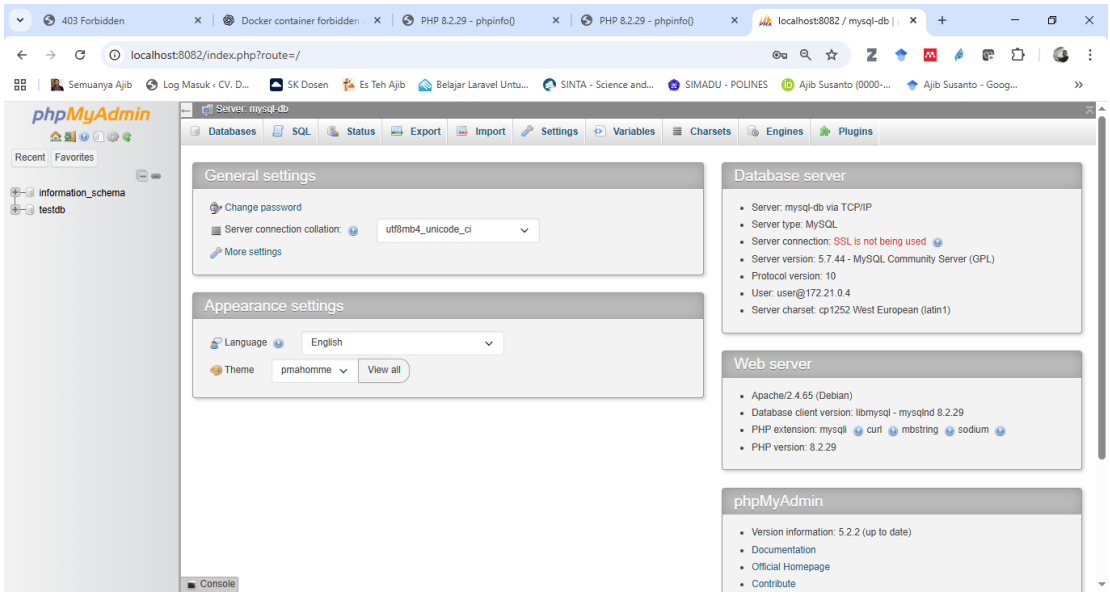
```
docker-compose up -d --build
```

g. Buka web :

- Web: http://localhost:8081
- phpMyAdmin: http://localhost:8082



System	Linux 3c8cd0c6869b 5.15.153.1-microsoft-standard-WSL2 #1 SMP Fri Mar 29 23:14:13 UTC 2024 x86_64
Build Date	Sep 30 2025 00:03:41
Build System	Linux - Docker
Build Provider	https://github.com/docker-library/php
Configure Command	'./configure' '--build=x86_64-linux-gnu' '--with-config-file-path=/usr/local/etc/php' '--with-config-file-scan-dir=/usr/local/etc/php/conf.d' '--enable-option-checking=fatal' '--with-mhash' '--with-pic' '--enable-mbstring' '--enable-mysqlnd' '--with-password-argon2' '--with-sodium=shared' '--with-pdo-sqlite=/usr' '--with-sqlite3=/usr' '--with-curl' '--with-iconv' '--with-openssl' '--with-readline' '--with-zlib' '--disable-phpdbg' '--with-pear' '--with-libdir=lib/x86_64-linux-gnu' '--disable-cgi' '--enable-fpm' '--with-fpm-user=www-data' '--with-fpm-group=www-data' 'build_alias=x86_64-linux-gnu'
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/usr/local/etc/php
Loaded Configuration File	(none)
Scan this dir for additional .ini files	/usr/local/etc/php/conf.d
Additional .ini files parsed	/usr/local/etc/php/conf.d/docker-fpm.ini, /usr/local/etc/php/conf.d/docker-php-ext-mysqli.ini, /usr/local/etc/php/conf.d/docker-php-ext-opcache.ini, /usr/local/etc/php/conf.d/docker-php-ext-pdo_mysql.ini, /usr/local/etc/php/conf.d/docker-php-ext-sodium.ini
PHP API	20220829
PHP Extension	20220829
Zend Extension	420220829
Zend Extension Build	API420220829,NTS
PHP Extension Build	API20220829,NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	provided by mbstring



h. Buat file `koneksi.php` di folder src :

```php
<?php
$servername = "mysql-db"; // GUNAKAN nama service dari docker-compose.yml
$username = "user";
$password = "password";
$dbname = "testdb";

$conn = mysqli_connect($servername, $username, $password, $dbname);

if (!$conn) {
    die("Failed to connect to MySQL: " . mysqli_connect_error());
}
echo "✅ Connected successfully to MySQL in Docker!";
?>
```

Rebuild semua container :

```
docker-compose down
docker-compose up -d --build
```

3. Deployment Wordpress dengan Database MySQL & Redis